



**FACULTY OF SCIENCE
DEPARTMENT OF MATHEMATICS
ACADEMIC YEAR 2024/25
FINAL EXAMINATION**

Course Code: MAT111	Subject: MATHEMATICS	
	Title of Paper: INTRODUCTORY MATHEMATICS I	
Semester: 1		
Time Allowed: 120 MINUTES	Reading Time: 10 MINUTES	Maximum Marks: <u>100</u>
Examiners: Dr. R. Tshelametse, Dr. T. G. Motsumi, Mr. B. Lephodisa, Dr. O. Dira, Mr. M. Moetele, Ms. S. Mogotsi, Prof. B. Muatjetjeja, Dr. F. Thabologo, Dr. M. Sangago, Mr. P. Moile & Ms. K. Matongo		
INSTRUCTIONS TO CANDIDATES 1. Answer all FOUR (4) questions in SECTION A . [60 Marks] 2. Answer any TWO (2) questions out of THREE (3) in SECTION B . [40 Marks] 3. Show your working, omission of essential working will result in loss of marks. 4. Always simplify and write your final answer to the simplest /or lowest/ form.		

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NUMBER OF PAGES INCLUDING THE COVER PAGE: 12

MARKING KEY

SECTION A

Answer all **FOUR QUESTIONS** in this section.

QUESTION A1

(a) Given the function

$$f(x) = \frac{4x + 1}{3x - 2}.$$

(i) Find the domain of f . [2]

Answer: Because

$$3x - 2 = 0 \quad \text{implies} \quad x = \frac{2}{3},$$

we have

The Domain of $f = \mathbb{R} \setminus \{\frac{2}{3}\}$

OR

The domain of f is the set of all real $x \neq \frac{2}{3}$.

(ii) Find the inverse function of f . [5]

Solution.

OPTION 1

$$\begin{aligned} f(x) &= \frac{4x + 1}{3x - 2} \\ y &= \frac{4x + 1}{3x - 2} \\ x &= \frac{4y + 1}{3y - 2} \\ x(3y - 2) &= 4y + 1 \\ y(3x - 4) &= 2x + 1 \\ y &= \frac{2x + 1}{3x - 4} \\ \therefore f^{-1}(x) &= \frac{2x + 1}{3x - 4} \end{aligned}$$

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OPTION 2

$$\begin{aligned} f(x) &= \frac{4x + 1}{3x - 2} \\ y &= \frac{4x + 1}{3x - 2} \\ y(3x - 2) &= 4x + 1 \\ x(3y - 4) &= 2y + 1 \\ x &= \frac{2y + 1}{3y - 4} \\ y &= \frac{2x + 1}{3x - 4} \\ \therefore f^{-1}(x) &= \frac{2x + 1}{3x - 4} \end{aligned}$$

- (iii) Evaluate $(f^{-1} \circ f)(x)$ for any x in the domain of f . [5]

Solution. For $x \neq \frac{2}{3}$,

$$\begin{aligned}(f^{-1} \circ f)(x) &= f^{-1}(f(x)) = \frac{2f(x) + 1}{3f(x) - 4} \\&= \frac{\frac{2(4x+1)}{3x-2} + 1}{\frac{3(4x+1)}{3x-2} - 4} \\&= \frac{\frac{8x+2+3x-2}{3x-2}}{\frac{12x+3-12x+8}{3x-2}} \\&= \frac{11x}{11} \\&= x\end{aligned}$$

- (b) Determine whether $f(x) = \frac{x^2 + 1}{3x^3 + x}$ is odd, even or neither. [3]

Solution. For $x \neq 0$, we have

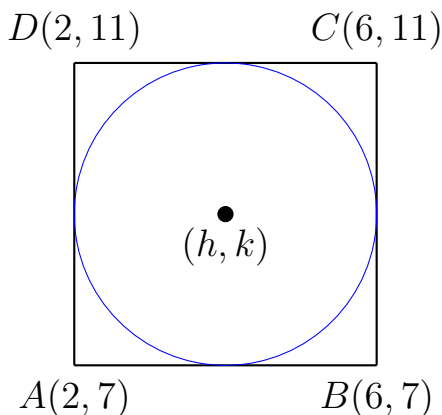
$$f(-x) = \frac{(-x)^2 + 1}{3(-x)^3 + (-x)} = \frac{x^2 + 1}{-3x^3 - x} = -\frac{x^2 + 1}{3x^3 + x} = -f(x).$$

Therefore, f is an odd function.

QUESTION A2

- (a) Find the standard form of the equation of the circle inscribed in the square whose vertices are at the points $A(2, 7)$, $B(6, 7)$, $C(6, 11)$ and $D(2, 11)$. [6]

Solution. The figure below shows the picture of the circle.



The mid-point of the diagonal $\overline{AC}$ (or the diagonal $\overline{BD}$) of the square is the center $O = (h, k)$ of the circle. Thus by the Mid-Point formula we have

$$O = (h, k) = \text{Mid-point of } \overline{AC} = \left(\frac{2+6}{2}, \frac{7+11}{2} \right) = \underline{\underline{(4, 9)}}$$

OR

$$O = (h, k) = \text{Mid-point of } \overline{BD} = \left(\frac{6+2}{2}, \frac{7+11}{2} \right) = \underline{\underline{(4, 9)}}.$$

The mid-points of the four sides $\overline{Ab}$, $\overline{BC}$, $\overline{CD}$ and $\overline{DA}$ of the square $ABCD$ lie on the circle. Let us take the midpoint of $\overline{AB}$.

$$E = \text{Mid-point of } \overline{AB} = \left(\frac{2+6}{2}, \frac{7+7}{2} \right) = \underline{\underline{(4, 7)}}.$$

Thus, by using the distance formula, the radius is given by

$$r = OE = \sqrt{(4-4)^2 + (9-7)^2} = \sqrt{4} = \underline{\underline{2}}.$$

Therefore, the standard form of the equation of the circle is given by

$$(x-h)^2 + (y-k)^2 = r^2$$

$$\boxed{(x-4)^2 + (y-9)^2 = 2^2 = 4}.$$

(b) Given the quadratic function $f(x) = x^2 - 3x + 1$.

(i) Write the quadratic function in standard form. [4]

Solution.

OPTION 1. By completing the square || OPTION 2. Using Formula

$$f(x) = x^2 - 3x + 1$$

$$|| f(x) = a(x-h)^2 + k$$

$$= (x^2 - 3x) + 1$$

$$|| h = -\frac{b}{2a} = -\frac{-3}{2} = \frac{3}{2}$$

$$= (x^2 - 3x + \frac{9}{4}) + 1 - \frac{9}{4}$$

$$|| k = c - \frac{b^2}{4a} = 1 - \frac{9}{4} = -\frac{5}{4}$$

$$= \left(x - \frac{3}{2} \right)^2 - \frac{5}{4}$$

$$|| \therefore f(x) = \left(x - \frac{3}{2} \right)^2 - \frac{5}{4}$$

(ii) Identify the vertex of the graph of f . [1]

Answer. The vertex is $\left(\frac{3}{2}, -\frac{5}{4} \right)$.

- (c) Find the value of B if $x - 1$ is a factor of the polynomial $P(x) = x^4 - 2x^3 + Bx^2 + 3x + 2$. [2]

Solution. By the Factor Theorem, $P(1) = 0$; that is,

$$0 = P(1) = 1 - 2 + B + 3 + 2$$

$$\underline{\underline{B = -4}}$$

- (d) Find the remainder when the polynomial $P(x) = x^3 + 8x^2 - 15$ is divided by $x + 2$. [2]

Solution. By the Remainder Theorem, $P(-2)$ is the remainder.

Thus

$$\underline{\underline{P(-2) = (-2)^3 + 8(-2)^2 - 15 = -8 + 32 - 15 = 9.}}$$

QUESTION A3

- (a) Evaluate $f(x) = \log_5 \left(\frac{2x}{x^2 - 1} \right)$ at $x = 4$. [2]

Answer. $f(4) = \log_5 \left(\frac{2(4)}{4^2 - 1} \right) = \log_5 \left(\frac{8}{15} \right) \approx -0.391$

- (b) Solve the following equations.

(i) $3^{2x} - 9^{1-x} = 0$ (ii) $\log_3 \left(\frac{x}{2x - 1} \right) = 2$ [3 + 3]

Solution. Using the definitions and rules of exponential and logarithmic functions, we have

$$(i) \quad 3^{2x} - 9^{1-x} = 0$$

$$3^{2x} = 9^{1-x}$$

$$3^{2x} = 3^{2(1-x)}$$

$$2x = 2 - 2x$$

$$x = \underline{\underline{\frac{1}{2}}}$$

$$(ii) \quad \log_3 \left(\frac{x}{2x - 1} \right) = 2$$

$$\frac{x}{2x - 1} = 3^2$$

$$x = 9(2x - 1)$$

$$-17x = -9$$

$$x = \underline{\underline{\frac{9}{17}}}$$

(c) Given the angle $\theta = -240^\circ$.

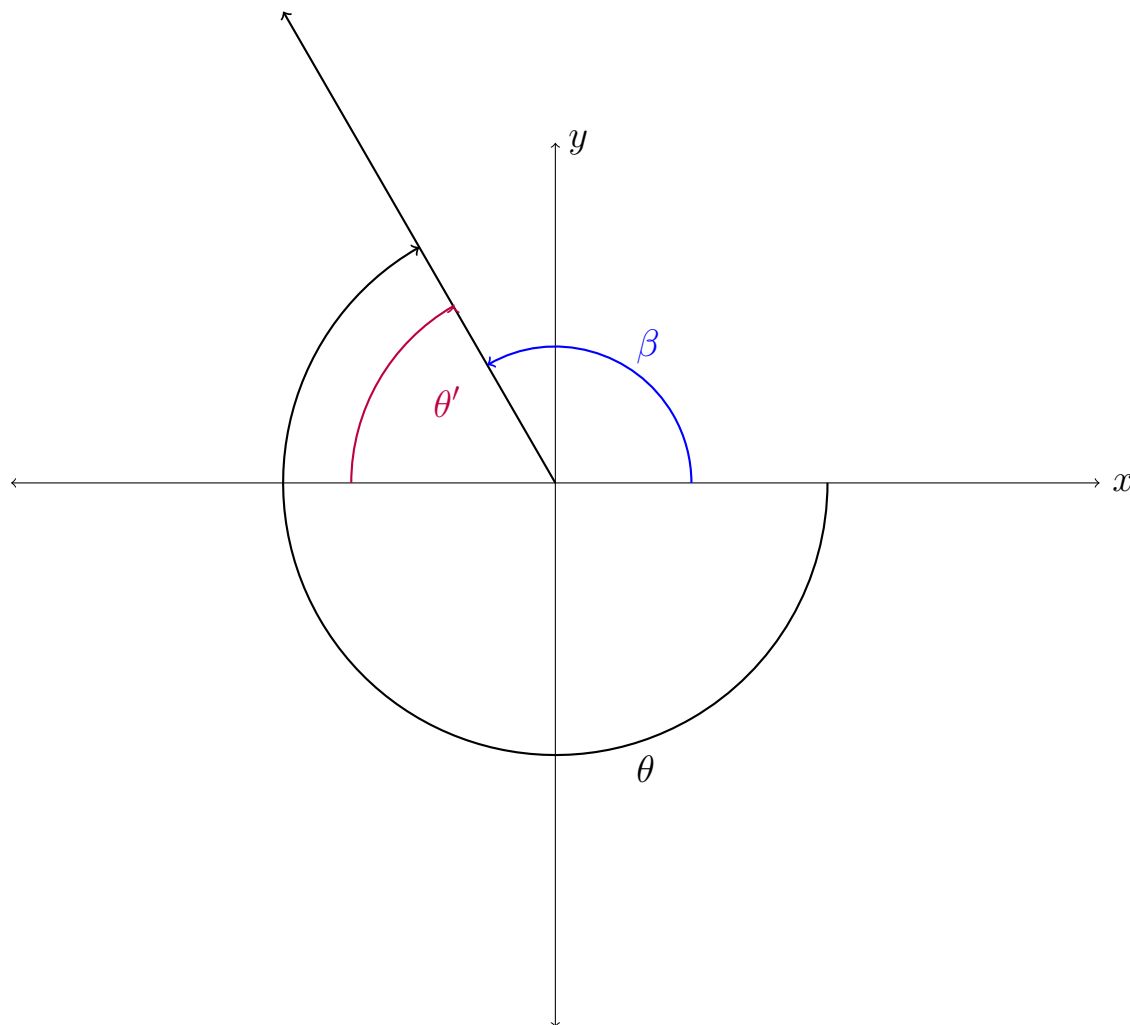
- (i) Find the reference angle θ' for the angle θ . Sketch θ in standard position and label θ' . [4]

Solution. Because θ is negative, $\theta = -240^\circ$ is coterminal with the angle

$$\beta = 360^\circ + \theta = 360^\circ - 240^\circ = 120^\circ$$

whose terminal side lies in Quadrant II. Therefore, the reference angle is given by

$$\theta' = 180^\circ - 120^\circ = 60^\circ.$$



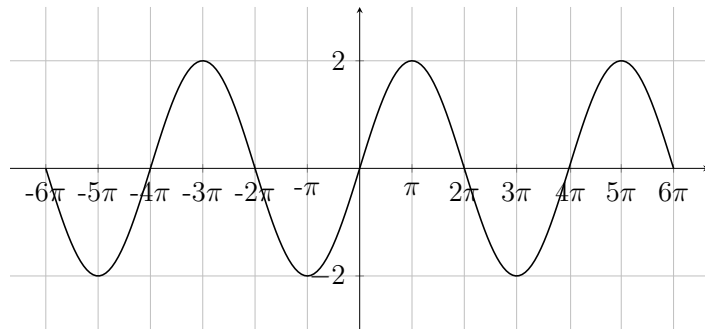
- (ii) Find the exact value of $\tan \theta$ without using a calculator. [3]

Solution. Because θ is a second quadrant angle, $\tan \theta$ is negative, and so

$$\tan \theta = -\tan \theta' = -\tan (60^\circ) = -\sqrt{3}.$$

QUESTION A4

- (a) Consider the graph of a periodic function given below.



- (i) Give the amplitude of the function. [1]

Answer. Amplitude= 2.

- (ii) Give the period of the function. [1]

Answer. Period= 4π .

- (iii) Give the equation of the function. [3]

Answer. $f(x) = 2 \sin\left(\frac{1}{2}x\right)$.

- (b) Prove the trigonometric identity $\frac{1}{1 - \sin x} + \frac{1}{1 + \sin x} = 2 \sec^2 x$. [5]

Solution. The right side is more complicated, so start with it.

$$\frac{1}{1 - \sin x} + \frac{1}{1 + \sin x} = \frac{1 + \sin x + 1 - \sin x}{(1 - \sin x)(1 + \sin x)} \quad \text{Add fractions.}$$

$$= \frac{2}{1 - \sin^2 x} \quad \text{Simplify.}$$

$$= \frac{2}{\cos^2 x} \quad \text{Pythagorean identity}$$

$$= 2 \sec^2 x \quad \text{Reciprocal identity}$$

- (c) Write the complex number $\frac{2i}{2+i} + \frac{5}{2-i}$ in standard form. [5]

Solution.

$$\begin{aligned} \frac{2i}{2+i} + \frac{5}{2-i} &= \frac{2i(2-i) + 5(2+i)}{(2+i)(2-i)} \\ &= \frac{4i - 2i^2 + 10 + 5i}{4 - 2i + 2i - i^2} \\ &= \frac{12 + 9i}{5} = \frac{12}{5} + \frac{9}{5}i \end{aligned}$$

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## SECTION B

Answer any TWO(2) out of THREE(3) Questions in this section.

### QUESTION B1

- (a) The graph of the function  $f(x) = |x|$  has gone through a series (one after the other) of the following transformations; first it was shifted 1 unit to the right, then reflected by about  $x$ -axis and finally shifted 2 units up. At each stage, give the equation of the 'new' function and sketch the graph of this 'new' function. [11]

**Solution.** Given  $f(x) = |x|$ .

\* Transformations

- First Transformation: Horizontal Shift 1 unit to the right

$$g(x) = f(x - 1) = |x - 1|.$$

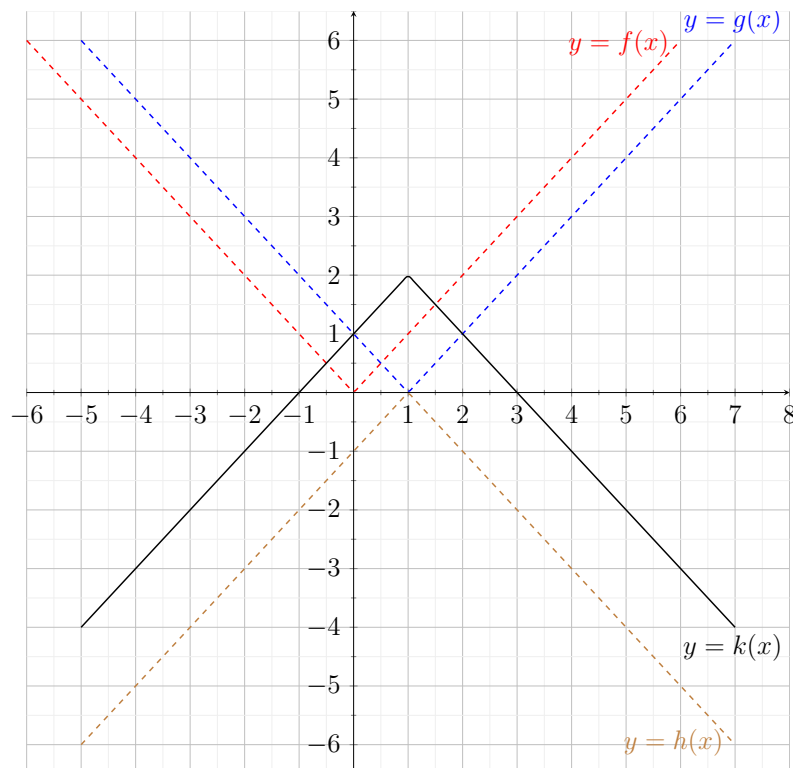
- Second Transformation: Reflection about  $x$ -axis

$$h(x) = -g(x) = -|x - 1|.$$

- Third Transformation: Vertical shift 2 units up

$$k(x) = h(x) + 2 = -|x - 1| + 2.$$

\* Graphs of  $f$ ,  $g$ ,  $h$  and  $k$





- (b) Find all the solution of  $\sec^2 x - \sec x - 2 = 0$  for  $x \in [0, 2\pi)$ . [9]

**Solution.**

$$\sec^2 x - \sec x - 2 = 0$$

Write original equation.

$$(\sec x - 2)(\sec x + 1) = 0$$

Factorize.

$$\sec x - 2 = 0$$

$$\sec x + 1 = 0$$

Each Factor equal zero.

$$\sec x = 2$$

$$\sec x = -1$$

Solve for  $\sec x$

We note that the equation  $\sec x = 2$  has solutions

$$x = \frac{\pi}{3} \quad \text{and} \quad x = \frac{5\pi}{3}$$

in the interval  $[0, 2\pi)$ . We also note that the equation  $\sec x = -1$  has exactly one solution

$$x = \pi$$

in the interval  $[0, 2\pi)$ . Therefore, the solutions of the equation in the interval  $[0, 2\pi)$  are

$$x = \frac{\pi}{3}, \quad x = \pi \quad \text{and} \quad x = \frac{5\pi}{3}.$$

## QUESTION B2

- (a) Given the function  $f(x) = -2 \tan(2x)$ .

- (i) Find the period of the function. [1]

**Answer.** Period =  $\frac{\pi}{2}$ .

- (ii) Find the  $x$ - and  $y$ - intercepts (if any) of the graph of  $f$  between  $-\frac{\pi}{4}$  and  $\frac{3\pi}{4}$ . [2]

**Answer.**

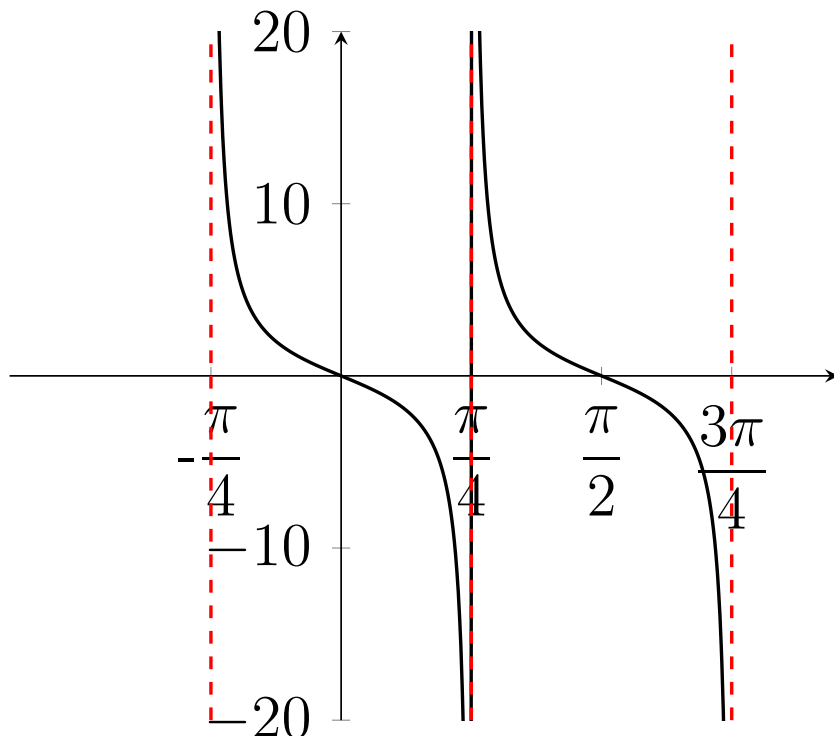
- $x$ - intercepts are  $(0, 0)$  and  $(\frac{\pi}{2}, 0)$ .
- $y$ - intercept is  $(0, 0)$ .

- (iii) Find vertical asymptotes (if any) of the graph of  $f$  between  $-\frac{\pi}{4}$  and  $\frac{3\pi}{4}$ . [2]

**Answer.** The vertical lines  $x = -\frac{\pi}{4}$ ,  $x = \frac{\pi}{4}$  and  $x = \frac{3\pi}{4}$  are the vertical asymptotes of the graph of  $f$ .

- (iv) Sketch the graph of the function between  $-\frac{\pi}{4}$  and  $\frac{3\pi}{4}$ . [5]

GRAPH OF  $f(x) = -2 \tan(2x)$



- (b) Use long division to find the quotient and remainder when the polynomial function  $P(x) = x^4 - 3x^3 + 6x^2 + 3x - 63$  is divided by the polynomial function  $d(x) = x^2 - 2x + 10$ . [10]

**Solution.**

$$\begin{array}{r}
 x^2 - 2x + 10 \overline{) \begin{array}{r} x^4 - 3x^3 + 6x^2 + 3x - 63 \\ - (x^4 + 2x^3 - 10x^2) \\ \hline -x^3 - 4x^2 + 3x \\ - (x^3 - 2x^2 + 10x) \\ \hline -6x^2 + 13x - 63 \\ - (6x^2 - 12x + 60) \\ \hline x - 3 \end{array} }
 \end{array}$$

$$P(x) = d(x)q(x) + r(x) \quad \text{where Quotient: } q(x) = x^2 - x - 6$$

$$\text{Remainder: } r(x) = (x - 3)$$

### QUESTION B3

(a) Given the quadratic function  $f(x) = -x^2 + 4x + 6$ .

(i) Find the  $x$ - and  $y$ - intercepts of the graph of  $f$ . [5]

**Solution.**

- $x$ -intercepts: We solve  $f(x) = 0$ ; that is solve the quadratic equation

$$-x^2 + 4x + 6 = 0.$$

OPTION 1: Extraction of Root || OPTION 2: Quadratic Formula

$$\begin{array}{ll} -x^2 + 4x + 6 = 0 & || \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ -(x - 2)^2 + 10 = 0 & || \quad a = -1, b = 4, c = 6 \\ (x - 2)^2 = 10 & || \quad x = \frac{-4 \pm \sqrt{16 + 24}}{-2} \\ x = \pm \sqrt{10} & || \quad x = 2 \pm \sqrt{10} \end{array}$$

Therefore, the  $x$ -intercepts are  $(2 - \sqrt{10}, 0)$  and  $(2 + \sqrt{10}, 0)$ .

- The  $y$ -intercept is  $(0, f(0)) = (0, 6)$ .

(ii) Find the coordinates of the vertex. [3]

**Solution.**

OPTION 1. By completing the square || OPTION 2. Using Formula

$$\begin{array}{ll} f(x) = -x^2 + 4x + 6 & || \quad f(x) = a(x - h)^2 + k \\ = -(x^2 - 4x) + 6 & || \quad h = -\frac{b}{2a} = -\frac{4}{-2} = 2 \\ = -(x^2 - 4x + 4) + 6 + 4 & || \quad k = c - \frac{b^2}{4a} = 6 - \frac{16}{-4} = 10 \\ = -(x - 2)^2 + 10 & || \quad \therefore f(x) = (x - 2)^2 + 10 \end{array}$$

Therefore, the vertex is  $V(2, 10)$ .

(iii) State whether the vertex is a maximum or minimum point of the graph. [1]

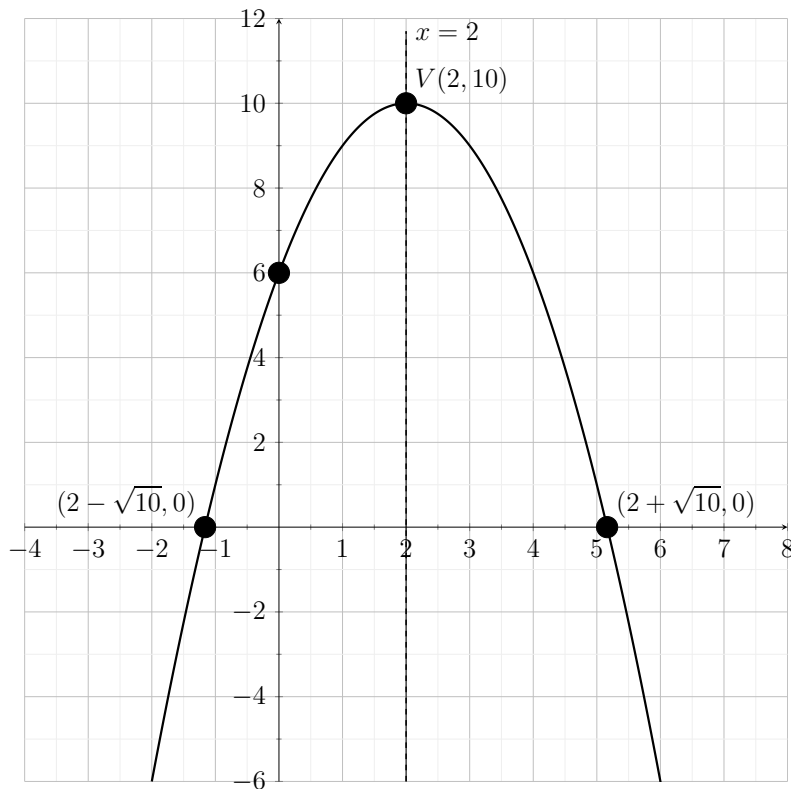
**Answer.** The vertex  $(2, 10)$  is the maximum point of the graph.

- (iv) Find axis of symmetry of the graph of  $f$ . [1]

**Answer.** Axis of symmetry is the vertical line  $x = 2$ .

- (v) Sketch the graph of  $f$ . [4]

**Solution.**



- (b) Solve  $\log_2 x + \log_2 (x - 3) = 2$ . [6]

**Solution.** We observe that the domain of the equation is  $(3, \infty)$ . We apply rules of logarithm to obtain

$$\log_2 x + \log_2 (x - 3) = 2$$

$$\log_2 x(x - 3) = 2$$

$$x(x - 3) = 2^2$$

$$x^2 - 3x - 4 = 0$$

$$(x - 4)(x + 1) = 0$$

$$x - 4 = 0 \Rightarrow x = 4$$

$$x + 1 = 0 \Rightarrow x = -1$$

Therefore,  $x = 4$  is the only solution of the given equation.

~~~~~ **END** ~~~~~